Psychiatry Research 289 (2020) 112960

Contents lists available at ScienceDirect

Psychiatry Research

journal homepage: www.elsevier.com/locate/psychres

Prediction of physical violence in schizophrenia with machine learning algorithms

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ARTICLE INFO

Keywords:
Violence
Schizophrenia
Childhood trauma
Personality
Machine learning

ABSTRACT

Patients with schizophrenia have been shown to have an increased risk for physical violence. While certain features have been identified as risk factors, it has been difficult to integrate these variables to identify violent patients. The present study thus attempts to develop a clinically-relevant predictive tool. In a population of 275 schizophrenia patients, we identified 103 participants as violent and 172 as non-violent through electronic medical documentation, and conducted cross-sectional assessments to identify demographic, clinical, and sociocultural variables. Using these predictors, we utilized seven machine learning classification algorithms to predict for past instances of physical violence. Our classification algorithms predicted with significant accuracy compared to random discrimination alone, and had varying degrees of predictive power, as described by various performance measures. We determined that the random forest model performed marginally better than other algorithms, with an accuracy of 62% and an area under the receiver operator characteristic curve (AUROC) of 0.63. To summarize, machine learning classification algorithms are becoming increasingly valuable, though, optimization of these models is needed to better complement diagnostic decisions regarding early interventional measures to predict instances of physical violence.

1. Introduction

The frequency of violence in patients with schizophrenia is disproportionately higher compared to that of the general population (Hodgins, 2008). From an epidemiological perspective, the one-year prevalence of violent behavior was 8.4% in schizophrenia patients, as opposed to 2.1% in persons without a history of psychiatric disorders (Swanson et al., 1990). While the exact proportion of violent patients is inconsistent, the robust association between violent behavior and schizophrenia has been repeatedly documented by numerous studies (Hodgins, 2008) (Tiihonen et al., 1997; Wallace et al., 2004). Such violent behaviors can result in not only physical injury or death to the assailant and victims, but also cause unnecessary emotional suffering for caregivers and loved ones (Volavka, 2013). Furthermore, violence and the fear of violence in the clinical setting could potentially complicate the treatment process by compromising the quality of patient care, affecting diagnostic decision making, and prolonging patient hospitalization (Hodgins, 2008; Iozzino et al., 2015)

An important consideration to note is that violent tendencies are expressed only in a minority of patients with schizophrenia (Hodgins, 2008). Many studies have attempted to explore factors that may increase the risk for development of violent behavior. Since violent behavior occurs within the realm of a patient with a unique life history, state of health, and social surroundings, it is important to collectively consider multiple factors (Swanson et al., 2006). In a recent systematic review and meta-regression analysis of 110 studies, Witt et al. found several risk factors for violence in psychosis, including being male, having a lower socioeconomic status, experiencing physical and sexual abuse during childhood, and having a history of substance abuse (Witt et al., 2013). **Furthermore, a meta-analysis of 32 meta-analyses showed an association between substance use and violence. They found male gender, psychotic disorders, combined alcohol/drug use increased the associations between substance abuse and violence (Duke et al., 2018).** However, despite the identification of risk factors that show statistically significant differences between violent and non-violent patients, it is imperative to utilize these findings to address the lack of efficient and valid risk assessment instruments to identify patients at risk for violent behavior (Silverstein et al., 2015). Machine learning classification algorithms can provide the critical link between identifying at-risk patients and administering preventive

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https://doi.org/10.1016/j.psychres.2020.112960
Received 22 October 2019; Received in revised form 17 March 2020; Accepted 27 March 2020
Available online 13 April 2020
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measures and treatment if necessary (Citrome, 2015).

Machine learning can be broadly defined as a computational technique that involves computers automatically learning from data inputs and delivering a solution (Deo, 2015; Dwyer et al., 2018). Classification algorithms belong to a subset of machine learning, and utilize predefined rules to assess various measurements and assign subjects to individual categories (Harper, 2005). With respect to psychiatry, it has been demonstrated that the use of classification algorithms is a highly effective method of incorporating various clinical measurements to predict patient outcomes. For instance, machine learning classifiers have been used to predict non-remitting post-traumatic stress disorder from clinical, psychophysiological, and demographic information (Galatzer-Levy et al., 2017). Furthermore, machine learning approaches have been used to predict suicide attempt in schizophrenia using clinical and sociocultural features (Hettige et al., 2017) and data from psychiatric scales and sociodemographic elements (Oh et al., 2017). Likewise, prediction of lifetime violence in schizophrenia can also be completed by machine learning classification algorithms.

Seven classification algorithms were implemented in the present study. While each classifier has unique methodologies in the processing of inputs and modeling complex relationships, it has been shown that no single algorithm performs consistently better than the others (Harper, 2005). Rather, the best performing algorithm is often dependent on the type of data and the features of the particular dataset that is to be analyzed (Harper, 2005). For this classification problem, predictor variables were based on demographic, clinical, and sociocultural factors that were previously suggested to be associated with violence in schizophrenia. Thus, the overall objective of the study was to create a clinically-relevant predictive model to identify patients with potential violent tendencies, and attempt to identify the classification algorithms most suited for this area of schizophrenia research.

2. Methods

A total of 275 participants were recruited from the Centre for Addiction and Mental Health (CAMH), located in Toronto, Canada. **All study participants met the inclusion criteria of being between the ages of 18 and 75; and the diagnosis of schizophrenia spectrum disorder was confirmed by the Structured Clinical Interview for the DSM-IV (SCID-I/P).** Exclusion criteria included having any diagnosis of an intellectual disability, major neurological disorder, or substance-induced psychosis; or having a history of head trauma resulting in loss of consciousness. All participants provided written informed consent for study participation and disclosure of personal health records for electronic medical chart verification. The study was approved by the CAMH Research Ethics Board.

The presence of violence was determined retrospectively through a review of all available patient electronic medical records for any documentation of incidences of physically violent behavior. We assigned severity scores for each participant ranging from an absence of physical violence to assault resulting in serious bodily injury, as outlined in the Modified Overt Aggression Scale (MOAS) (Kay et al., 1988). For the purpose of this study, participants were simply classified as violent or non-violent. To predict violence with machine learning algorithms, we incorporated 28 predictive variables based on previously identified factors associated with violence or schizophrenia. These variables were obtained by administering cross-sectional assessments in the form of structured interviews and self-report questionnaires, and utilizing psychiatric rating scales and questions assessing medical history and demographic information.

Relevant risk factors that were incorporated included age, sex, age-of-onset of psychosis, and number of previous psychiatric hospitalizations. The SCID-I/P assessed for comorbid diagnoses of lifetime alcohol, drug, and marijuana abuse or dependence, whereas the Family Interview for Genetics Studies (FIGS) identified family histories of psychosis, mood disorders, and suicide (Gershon and Guroff, 1984). Suicidal behaviors are significantly associated with violence in schizophrenia (Witt et al., 2013), and as such, each participant's history of suicide attempt was determined through the Columbia- Suicide Severity Rating Scale (C-SSRS) (Posner et al., 2011). Immigration has also been shown to be an important risk factor for schizophrenia (Bourque et al., 2011; Cantor-Graae and Selten, 2005), and accordingly, aspects

Table 1
Descriptive statistics of demographic, clinical, and sociocultural predictor variables, with p-values. Independent *t*-test for numeric variables, and chi-squared test with Yates' correction for categorical variables.

Total (N = 275)	Violent (N = 103)	Non-Violent (N = 172)	P-Value
Age (μ ± s.d.)	44.82 ± 12.95	38.54 ± 12.83	< 0.001
Sex (male/ female)	81/22	116/56	0.064
European caucasian	63 (61%)	107 (62%)	0.965
English as primary language	85 (83%)	148 (86%)	0.540
Religious identity	80 (78%)	124 (72%)	0.379
Immigrated before age 9	15 (15%)	19 (11%)	0.504
Immigrated between age 10 and 18	18 (17%)	17 (10%)	0.101
Immigrated after age 18	2 (2%)	16 (9%)	0.033
Age-of-onset (μ ± s.d.)	23.34 ± 7.51	22.65 ± 7.41	0.460
Number of hospitalizations (N > 5)	52 (50%)	49 (28%)	< 0.001
Suicide attempter	41 (40%)	65 (38%)	0.838
Lifetime alcohol abuse/ dependence	38 (37%)	45 (26%)	0.082
Lifetime marijuana abuse/ dependence	27 (26%)	50 (29%)	0.710
Lifetime drug abuse/ dependence	37 (36%)	55 (32%)	0.590
Family history of psychosis	25 (24%)	49 (28%)	0.534
Family history of mood disorders	34 (33%)	79 (46%)	0.048
Family history of suicidal behavior	25 (24%)	39 (23%)	0.876
NEO neuroticism (μ ± $s.d.$)	33.75 ± 8.64	34.14 ± 9.55	0.734
NEO extraversion (μ ± $s.d$)	36.98 ± 6.61	37.65 ± 7.11	0.436
NEO openness (μ ± $s.d$)	38.42 ± 6.16	39.93 ± 6.73	0.065
NEO agreeableness (μ ± $s.d$)	39.68 ± 5.36	41.26 ± 5.04	0.015
NEO conscientiousness (μ ± $s.d$)	42.60 ± 7.20	41.86 ± 8.09	0.442
CTQ physical abuse (μ ± $s.d$)	8.63 ± 4.92	7.63 ± 3.93	0.067
CTQ emotional abuse (μ ± $s.d$)	10.68 ± 5.26	10.79 ± 5.41	0.881
CTQ sexual abuse (μ ± $s.d$)	8.33 ± 4.92	7.43 ± 4.62	0.125
CTQ physical neglect (μ ± $s.d$)	9.27 ± 3.80	8.31 ± 3.75	0.042
CTQ emotional neglect (μ ± $s.d$)	12.46 ± 4.98	11.37 ± 5.42	0.098
CTQ total score (μ ± $s.d.$)	44.48 ± 18.44	41.26 ± 16.20	0.130

regarding self-reported ethnicity, primary language, religious identity, and age of immigration were recorded. Furthermore, both childhood trauma and five-factor personality traits have been linked with violent behavior in psychosis (Lust, 2017; Weiler, 1996), and therefore, data from the Childhood Trauma Questionnaire (CTQ) (Bernstein et al., 1994) and NEO Five Factor Inventory (NEO-FFI) (Costa, 1992) were also included. Table 1 provides a descriptive summary of all predictive variables utilized in the study. We also performed independent *t*-tests for all numeric variables and chi-squared tests with Yates' correction for all categorical variables, in which violence status was the outcome variable.

In order to assess and compare the validity of different machine learning models, we used several classification methods to identify participants as either violent or non-violent based on the predictive variables. We first ran a binary logistic regression model, the most popular multivariable statistical method in the field of health sciences (Park, 2013). Afterwards, we utilized six additional classification algorithms for model prediction comparison: the least absolute shrinkage and selection operator (lasso), elastic net, random forest, gradient boosted regression trees (GBRT), support vector machine (SVM) classifier, and SVM classifiers with radial basis function (RBF) kernels. All models were run on the Specialized Computing Cluster (SCC) at CAMH using the scikit-learn package in Python 2.7.

The lasso and elastic net algorithms build off the traditional regression model. The lasso implements regularization methods to optimize predictor selection and regression coefficient estimation by utilizing a penalization step of shrinking some predictor coefficients while setting some others to zero (McNeish, 2015; Tibshirani, 1996). This has shown improved predictive performance through reducing model overfitting (Musoro et al., 2014). The elastic net, a recently proposed algorithm, further incorporates a grouping effect where strongly correlated predictor variables are together in or out of the model (Zou, 2005). Studies have found that the elastic net algorithm can be especially useful in classification problems with a large number of predictor variables (Zou, 2005).

The random forest method can be described as an "ensemble learning" technique, in that it combines the results of multiple decision trees to return a single output or prediction (Breiman, 2001; Couronne et al., 2018). The individual trees of the metaphoric forest are decision tree classifiers built using a randomly selected sample of the original dataset. Furthermore, for each decision tree to make a prediction, it considers only a randomly selected group of predictor variables to establish its branching nodes. Ultimately, each tree casts a single vote, and the random forest algorithm, as a whole, selects the prediction with the most votes. Due to the random selection process described, it is expected that the classification results would vary from run to run. **As such, we repeated the random forest algorithm for 25 runs, and report the 25-run average.** A related ensemble technique is the GBRT. The major difference with GBRT is the strategy of ensemble formation- unlike random forest where the individual decision trees are created simultaneously, GBRT adds new regression trees sequentially, training new models based on the errors already learned (Natekin and Knoll, 2013).

SVM is an advanced machine learning method that excels at recognizing subtle patterns in complex datasets (Huang et al., 2018). SVM-built classifiers treat each sample to be classified as a point in a high-dimensional space, and create a decision boundary, or hyperplane, between the prediction classes (Noble, 2006). The classifiers then extrapolate class predictions based on the hyperplane. While the classic SVM classifier was meant to model linear relationships, kernel methods allow the classifiers to create higher-dimensional models for more complex datasets. The RBF kernel projects the original data into infinite dimensions, thus allowing its linear separation, much like the linear SVM classifier (Nanda et al., 2018).

While these classification algorithms can be powerful predictor tools, it is important to take into consideration the class imbalance that exists between the violent minority of schizophrenia patients and the majority without violent tendencies (Dubey et al., 2014). Thus, to preserve the ratio between violent and non-violent subjects, we used a stratified *k*-fold cross validation technique, assigning $k = 5$. Through this method, the sample population was divided into five equally-sized subgroups, or folds, while maintaining the ratio of class distribution (Hettige et al., 2017). Five rounds of training and validation were performed, such that each fold was used once as a validation set while the remaining four folds were used for training purposes (Hettige et al., 2017).

Measures of algorithm performance include the overall predictive accuracy, area under the receiver operator characteristic (ROC) curve (AUROC), sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV). The overall predictive accuracy is the proportion of correct predictions from all predictions made. The ROC curve, a plot of sensitivity versus specificity across varying cut-offs, was used to visually evaluate the overall performance of each classification algorithm, while the AUROC combines sensitivity and specificity measures to describe algorithms' inherent validity (Bradley, 1997; Hajian-Tilaki, 2013). Sensitivity is defined as the ratio of correctly predicted positives (violent subjects) among the population of violent subjects; likewise, specificity is defined as the ratio of correctly predicted negatives (non-violent subjects) among the population of non-violent subjects. The PPV is defined as the proportion of correctly predicted positives (violent subjects) among the total number of violent predictions made by the classification algorithm, whereas the NPV is defined as the proportion of correctly predicted negatives (non-violent subjects) among the total number of non-violent predictions made by the algorithm.

A chi-squared test was used to assess each model's prediction accuracy (correct predictions versus incorrect predictions) compared to predictions made based on random chance. Furthermore, the McNemar's test, a statistical test used to analyze matched pairs of data, was used to compare the prediction accuracy between the seven different classification algorithms (Adedokun, 2012).

3. Results

In our final study sample of 275 patients with schizophrenia, there were 103 (37%) who had histories of violence and 172 (63%) without any violent tendencies. Table 1 shows the descriptive statistics pertaining to the demographic, clinical, and sociocultural predictor variables utilized in the classification algorithms. We obtained *p*-values for each predictor variable, identifying significant differences between the violent and non-violent groups.

Performance measures (accuracy, AUROC, sensitivity, specificity, PPV, and NPV) were reported as the mean and standard error of the five folds, whereas for random forest, they were reported as the mean and standard error of the 25 individual runs. The accuracy, sensitivity, specificity, PPV, and NPV were computed from each algorithm's predicted violence statuses of participants compared to their true statuses. Summary confusion matrices for each algorithm's prediction are found in **Supplementary Figure S1**. The AUROC was determined from the ROC curves (**Supplementary Figure S2**). All seven classification algorithms were able to differentiate between violent and non-violent subjects with varying levels of predictive power, as described by the various performance measures.

The logistic regression model performed with an overall predictive accuracy of 0.57 ± 0.050 and an AUROC of 0.64 ± 0.066. Lasso had an accuracy of 0.60 ± 0.053 and an AUROC of 0.64 ± 0.065. Elastic net had an accuracy of 0.59 ± 0.05 and an AUROC of 0.64 ± 0.06. **Random forest had an accuracy of 0.62 ± 0.004 and an AUROC of 0.63 ± 0.005 over the 25 individual runs.** GBRT had an accuracy of 0.60 ± 0.037 and an AUROC of 0.63 ± 0.050. SVM had an accuracy of 0.57 ± 0.058 and an AUROC of 0.64 ± 0.067. Lastly, RBF-SVM had an accuracy of 0.60 ± 0.046 and an AUROC of 0.62 ± 0.048.

Table 2
Comparison of machine learning algorithm performance in predicting violence. Values provided as means and standard errors. *P*-values calculated from a chi-square test assessing model accuracy (correct versus incorrect predictions) compared to predictions made according to random chance alone.

	Logistic Regression	Lasso	Elastic Net	Random Forests	GBRT	SVM	RBF-SVM
Accuracy	0.57 ± 0.050	0.60 ± 0.053	0.59 ± 0.05	**0.62 ± 0.004**	0.60 ± 0.037	0.57 ± 0.058	0.60 ± 0.046
AUROC	0.64 ± 0.066	0.64 ± 0.065	0.64 ± 0.06	**0.63 ± 0.005**	0.63 ± 0.050	0.64 ± 0.067	0.62 ± 0.048
Sensitivity	0.54 ± 0.065	0.57 ± 0.062	0.59 ± 0.059	**0.32 ± 0.008**	0.22 ± 0.046	0.58 ± 0.082	0.45 ± 0.046
Specificity	0.59 ± 0.044	0.62 ± 0.055	0.59 ± 0.041	**0.80 ± 0.004**	0.82 ± 0.070	0.56 ± 0.049	0.69 ± 0.052
PPV	0.45 ± 0.057	0.48 ± 0.061	0.59 ± 0.047	**0.62 ± 0.008**	0.49 ± 0.079	0.44 ± 0.062	0.48 ± 0.067
NPV	0.68 ± 0.044	0.71 ± 0.045	0.58 ± 0.042	**0.54 ± 0.003**	0.64 ± 0.017	0.69 ± 0.055	0.67 ± 0.031
***P*-Value**	0.013	< 0.001	0.003	**<0.001**	0.001	0.026	< 0.001

Table 3
McNemar's test comparing the different classification algorithm predictive accuracy (correct versus incorrect predictions). Values above the diagonal represent p-values; those below the diagonal represent chi-square values.

	Logistic Regression	Lasso	Elastic Net	Random Forests	GBRT	SVM	RBF-SVM
Logistic regression	- - -	0.349	0.394	**0.144**	0.598	0.845	0.505
Lasso	0.878	- - -	0.639	**0.473**	1.000	0.233	0.917
Elastic NET	0.727	0.220	- - -	**0.292**	0.831	0.289	0.739
Random forests	**2.130**	**0.516**	**1.111**	**- - -**	**0.238**	**0.110**	**0.437**
GBRT	0.278	0.000	0.045	**1.391**	- - -	0.475	1.000
SVM	0.038	1.422	1.125	**2.560**	0.510	- - -	0.374
RBF-SVM	0.444	0.011	0.111	**0.605**	0.000	0.790	- - -

Table 2 summarizes all computed performance measures utilized in this study.

The chi-squared test showed that all classification models performed significantly better than predictions determined by random chance. The McNemar's test showed that the seven classification models did not differ significantly from each other in a paired comparison of correct versus incorrect predictions (Table 3).

4. Discussion

There have been numerous previous studies and reviews aimed at identifying risk factors for violence in patients with schizophrenia and other psychoses. To briefly illustrate several key findings, meta-analyses found that factors such as prior victimization, substance abuse comorbidity, previous suicide attempts, age, gender, and ethnicity are associated with violence (Fazel et al., 2009; Witt et al., 2013). Other works have linked childhood trauma and certain personality traits with violent behavior in psychosis (Lust, 2017; Weiler, 1996). However, even after recognizing many risk factors, it has proven difficult to incorporate them into an efficient clinically-relevant tool to identify patients at risk for violent behavior. Many of the existing structured clinical risk assessment tools are not applicable to all patients, are time-consuming, and may frequently involve clinician bias (Monahan, 1984). Therefore, efforts are aiming to develop models to automatically predict for violence with existing patient data.

Wootton et al. utilized several clinical predictor variables (age, sex, history of violence, drug use, and diagnosis of personality disorder) in a logistic regression model to identify patients with psychosis who would commit assaults in the community over the following two years (Wootton et al., 2008). They found that their models had a predictive accuracy comparable to existing risk assessment tools (Wootton et al., 2008). As an extension to their study, we proposed the use of a more comprehensive list of predictor variables. Here, we incorporated a total of 28 demographic, clinical, and sociocultural factors, similar to those used in a previous study by our group predicting suicide attempts in schizophrenia (Hettige et al., 2017). All variables in this analysis were either associated with violence in psychoses or with the development of schizophrenia. Furthermore, we utilized the logistic regression model, as well as six additional advanced machine learning classification algorithms. This study is one of the first to use such a wide variety of predictor variables and classification algorithms to predict violence in patients with schizophrenia.

Regarding the performance measures for each algorithm, as outlined in Table 2, it is imperative to take note to ensure the validity of these values in all machine learning models. There are several potential underlying factors to consider, such as class imbalance, as previously mentioned, and model overfitting. Overfitting refers to instances in which the classification model learns the intricacies unique to the training set, and in doing so, becomes unreliable when used on another set of data from the remaining study population (Handelman et al., 2019). Therefore, each classifier was run according to a stratified five-fold cross validation technique. This methodology addresses the class imbalance problem by maintaining the ratio of violent versus non-violent patients in each fold. Running five folds of each classifier as cross validation further increases prediction performance, improves the generalizability of results, and avoids model overfitting (Arlot, 2010; Handelman et al., 2019).

Our classification algorithms demonstrated an ability to identify violent and non-violent patients with schizophrenia with a predictive accuracy ranging from 57% to 62%. These levels of accuracy were shown to be greater than that based on random chance alone (Table 2). While overall predictive accuracy may intuitively indicate the performance of a classifier, the value of the AUROC is of greater significance. The ROC curve is a plot of sensitivity versus specificity, and plays an important role in determining the diagnostic ability of classifiers to identify the true state of study subjects (Hajian-Tilaki, 2013). The AUROC can be formally defined as the area representing the probability that a randomly chosen positive (violent subject) is correctly identified with greater suspicion than a randomly chosen negative (non-violent) subject (Bradley, 1997). Taking into account the component features of sensitivity and specificity, the AUROC is said to be a superior performance measure than overall predictive accuracy. Our classification algorithms had AUROC values ranging from 0.62 to 0.64.

Sensitivity has been previously mathematically defined in terms of a measure of classification performance. From a clinical perspective, however, sensitivity is the ability of a test to correctly identify patients with a particular disease- in our case, the identification of violent patients with schizophrenia (Lalkhen, 2008). In other fields of medicine, sensitivity is known to be especially useful in identifying a serious, yet treatable disease (Lalkhen, 2008). As such, an argument can be made for violence in patients with schizophrenia. Violent tendencies, both in and out of the hospital, can lead to serious consequences for everyone

involved, yet, with certain interventions, that risk may be mitigated. Treatment strategies in the management of violence often incorporate behavior plans and/or psychopharmacology (Fetter, 2011). Behavioral plans can include changes in the way clinicians converse with patients, and the addition of structured plans to emphasize positive reinforcement of adaptive behavior (Fetter, 2011). Psychopharmacology involves the prescription of various medications, such as long-acting injectable antipsychotics or clozapine (Fetter, 2011). It is important to mention that behavioral strategies are not necessarily detrimental to patients misclassified as violent, and that the prescription of certain antipsychotics is not entirely based on the violent tendencies. Therefore, in our study, sensitivity is considered to be one of the most important performance measures.

The other measures- specificity, PPV, and NPV- are additional values we can use to assess model performance, though they are less directly clinically-relevant in this study. Specificity refers to the ability to correctly identify patients without a given disease- in our case, the identification of non-violent patients (Lalkhen, 2008). Since there is a class imbalance with the majority of patients with schizophrenia not having violent tendencies, it is easy for even a poor classification algorithm to demonstrate a high specificity. In a theoretical example, if a primitive model assigned all subjects non-violent on the basis that the majority of patients are non-violent, the model would have an incredibly high specificity, but would not have any predictive power to address the original purpose of the model. On the other hand, this is not to say that specificity is in no way applicable to this study. The AUROC incorporates both sensitivity and specificity, and it is the measure of overall model performance. PPV refers to the likelihood that a patient has a given disease given a positive result, whereas NPV refers to the likelihood that a patient does not have a given disease given a negative result (Lalkhen, 2008). These values are dependent on the prevalence of the disease- in our case, the prevalence of violence- in a given population. While PPV and NPV are used to assess the value of a classification model to a clinician, it does not align well with the aims of developing a classification model with a primary emphasis on the identification of patients with violence.

Considering the thorough discussion of the various performance measures reported in this study, we found that the random forest algorithm using 25 runs performed better than the other models. Elastic net resulted in an overall accuracy of 59%, though the AUROC was the highest among all other models with a value of 64%. Furthermore, the sensitivity was also found to be the highest, with elastic net correctly classifying 59% of the violent patients in our study sample.

We used the McNemar's test for a paired comparison of model prediction accuracy. Interestingly, our results indicated that there were no significant differences in all possible pairs between the seven classification models.

We then compared the individual performance measures of random forest with each of SVM, logistic regression, and GBRT. While the accuracy of random forest was higher, the AUROC was comparable to the other three classification models and the sensitivity fell below that of logistic regression and SVM. Overall, the results from this study indicate that although random forest had almost significantly better prediction accuracy, it is a poor clinical model for identifying schizophrenia patients with potential for violence. On the other hand, the elastic net model performed marginally better than the other models in terms of sensitivity.

Compared to the results of other studies using different methods, we found approximately the same average level of sensitivity for our algorithms (Table 2). For example, in a clinical prediction study of Assaultive Behavior among male psychiatric patients by Hoptman et al. (Hoptman et al., 1999), the diagnostic sensitivity was 54%, while the sensitivity of our algorithms ranges between 22% (GBRT) and 59% (Elastic Net).

However, compared to the results from Wootton et al., our elastic net algorithm showed lower AUROC values- 0.64 versus 0.73 (Wootton et al., 2008).

When considering the single variables as predictors, surprisingly, we did not find a significant difference in rates of substance use disorders in our two groups (Table 1), although the link between substance use disorders and violence has been well established. However, since all participants were followed clinically in the study, if substance abuse emerged, it would have been dealt with swiftly. Also, our sample was not a forensic sample, where elevated rates of substance misuse would be expected in participants.

This study certainly had limitations in preventing higher AUROC values, sensitivity, and other performance measures. We attribute these to factors including having a relatively small sample size to train and validate our classification models. Generally speaking, machine learning models require greater sample sizes to effectively train and validate predictions.

Furthermore, the 'one in ten rule' states that one predictive variable can be studied for every ten events when doing regression analysis (Harrell et al., 1984; Peduzzi et al., 1996), the relatively low number of participants to variables was another limitation in this study.

Another limitation is that we were unable to use previous history of violence as a predictor variable like that used in Wootton et al. (Wootton et al., 2008). While previous history would have been an excellent predictor variable for future violence, the aim of our study was to use machine learning models to determine violent and non-violent status, as a risk assessment tool in the clinical setting. It would not have been feasible to follow participants over the course of their lifetime to determine the outcome violent status.

Also, one methodological limitation of this study is that we did not use a standardized instrument that has been used for predicting of violence in several studies such as the Historical, Clinical and Risk Management Scales (HCR–20) (Webster et al., 1997). The HCR-20 is a structured professional judgment approach to assessing risk of violence. It comprises 10 historical items, 10 clinical items, and 5 risk items. Although the HCR-20 was originally developed for forensic samples, it has also been used for civil patients. For example, among civilly committed inpatients (58% with a diagnoses of schizophrenia) large effects on accuracy of violence prediction by using the HCR-20 were found for violence occurring in the hospital or community (Douglas et al., 1999). In a recent study by O'Shea et al. (O'Shea et al., 2015), it was found that HCR-20 total score was a significant predictor of any aggression; however, the area under the curve values did not reach the threshold for a large effect size (O'Shea et al., 2015).

To conclude, we report the outcomes of seven classification algorithms used in predicting violence status in patients with schizophrenia, using various demographic, clinical, and sociocultural predictor variables. While at the moment, only a few of the models show clinical utility with respect to this particular area of psychiatric research, with the addition of other predictor variables such as genetic and neuroimaging data, as well as increased sample size training and validation, machine learning shows optimism in the field.

With further work, classification algorithms may have the ability to complement diagnostic decisions to improve the treatment and well-being of patients with schizophrenia. **However, we would caution that decisions regarding restrictions of liberty as a result for propensity for violence should never rely solely upon risk assessment tools. Patients' behavior, clinical status, willingness to take medications, response to interventions, and expressions of engaging in violence must all be taken into consideration.**

5. Financial support

This research was supported by a CIHR operating grant (MOP-119332) and the Miner's Lamp Innovation Fund in Prevention and Early

Detection of Severe Mental Illness at the University of Toronto (VDL). KZW is supported by an Ontario Graduate Scholarship and the Cleghorn Fellowship in Schizophrenia Research at the University of Toronto.

Declaration of Competing Interest

Authors declare no conflicts of interest.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.psychres.2020.112960.

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